Torsional resistance

LAMMPS dem13-twist-sds | Torque and spin decay | 28 October 2026

PASS	All four runs pass the official 1% spin check at 30 ms.

Official spin check at 30 ms

Run	dt (us)	Spin at 30 ms	Error (%)	Spin at 50 ms	1%
CPU dt	1	8.79993010	0.00079	8.00006971	PASS
GPU dt	1	8.79993010	0.00079	8.00006971	PASS
CPU dt/2	0.5	8.79993403	0.00075	8.00005892	PASS
GPU dt/2	0.5	8.79993403	0.00075	8.00005892	PASS

Spin is omega_z in rad/s, about the contact normal. The official twist_decay checker tests 8.8 rad/s at cumulative step 30000 with 1% relative tolerance. The ideal 50 ms endpoint 8.0 is supplementary.

Inputs and dimensional mapping

Quantity	Official input / FunDEM configuration
Sphere / initial state	Diameter 0.005 m; density 2700 kg/m^3; z0 = 0.0024999998 m; v = 0; omega_z = 10 rad/s
Wall / gravity	Plane z = 0; g = 1 m/s^2 downward; periodic x/y immaterial for the isolated stationary sphere
Normal / sliding	Hooke kn = 1e6 N/m; ks = 2.8e5 N/m; sliding friction = 0.5; e = 1; no normal/sliding/rolling/global damping
Official twisting SDS	Ktw = 1 N m/rad; Ctw = 1e-6 N m s/rad; mut = 1e-4 m; rolling friction = 0
FunDEM mapping	D = 2R = 0.005 m; ktorsion = Ktw/D^2 = 40000 N/m; ctorsion = Ctw/D^2 = 0.04 N s/m; torsional friction = mut/D = 0.02
Stiffness / damping mode	Explicit linear stiffness and explicit contact damping; E / Poisson ratio are inactive schema placeholders
Time / saved result	50 ms; dt = 1 / 0.5 us; 50000 / 100000 steps; native frames every 10 us; every-step CSV

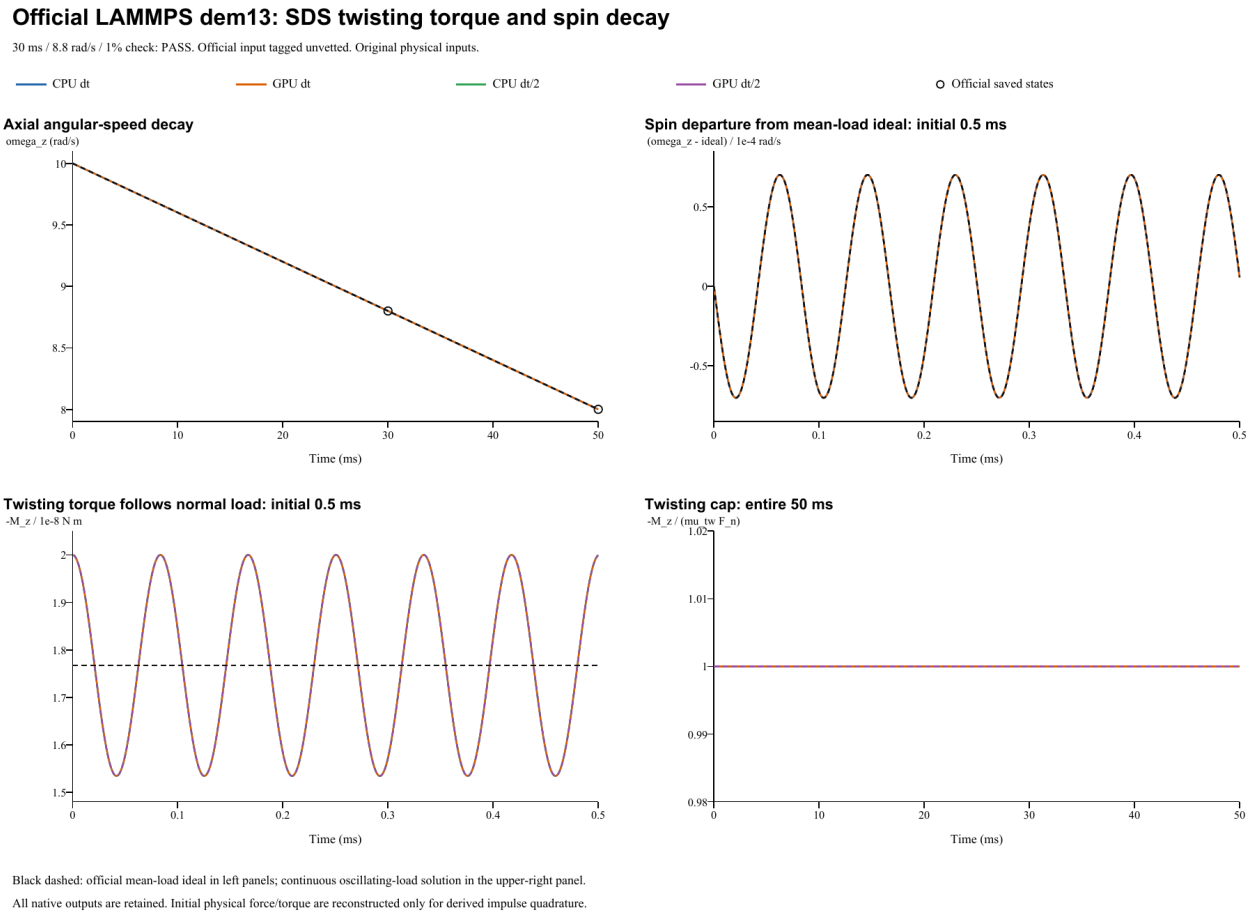
Analytic torque and spin

I = (2/5) m R^2 = 4.417864669e-10 kg m^2.
In the saturated regime: Mz = -mut Fn. With Fn = mg,
torque magnitude = 1.767145868e-08 N m; omega_z(t) = 10 - 40 t.

Reference provenance

Official YAML and golden states; analytic checker; SDS twisting model. Source tag: **granular unvetted**. Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; Git blob a6b317c73f34232186ceb3207a9422c02686f657. This newer pin is separate from earlier tests. LAMMPS was not rerun locally; epsilon 1e-10 is a regression setting.

Curves and retained evidence



Instantaneous torque cap: 1.5343e-08 to 2e-08 N m. CPU/GPU maximum spin difference: 0 rad/s; torque difference: 3.31e-24 N m. Maximum cap-law error / ideal torque: 1.87e-16; angular-momentum residual: 2.99e-22 N m s. Halving dt changes the 30 ms spin by 3.93e-06 rad/s. Maximum spin error against the oscillating-load solution changes from 6.01e-05 to 1.55e-05 rad/s.

Zero normal damping leaves +/-13.177% load oscillations; torque follows instantaneous Fn. The saved t = 0 state precedes force assembly. Impulse integration reconstructs Fn0 = 2e-4 N and Mz0 = -mut Fn0 without changing raw data. Contact force_z excludes gravity. Harmonic and impulse diagnostics supplement the official 30 ms target.

Complete native results are retained

CPU dt: run.json; 50,000 steps / 5,001 frames.

GPU dt: run.json; 50,000 steps / 5,001 frames.

CPU dt/2: run.json; 100,000 steps / 5,001 frames.

GPU dt/2: run.json; 100,000 steps / 5,001 frames.

Run manifest | Verified analysis | Checkpoint CSV | Contact audit | Full native ZIP. Native run.json, project.json, all 20,004 frames/contacts, input projects, backend logs, audits and every-step CSVs are kept.

Native output every 10 us includes both endpoints: 5001 frames per run, every 10 base or 20 half steps. CSVs separately retain every integration step.

Execution and validation limits

Real FunDEM SimulationSession runs use linked 0.4.3 CPU/CUDA libraries without solver changes or fitted parameters. GUI executable byte identity was not established.

This saturated, positive-spin test verifies the torsional cap and decay. General spring/dashpot limiting differs between the codes; uncapped response, damping sharing, reversals, sticking and cap transitions are not fully tested. Marshall torsion and many-particle flow remain outside scope.